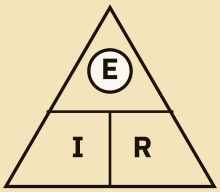
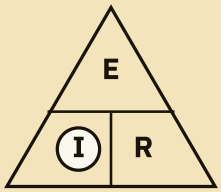
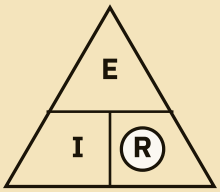
TECHNICIAN · T5

Formula card

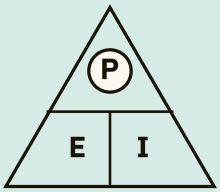
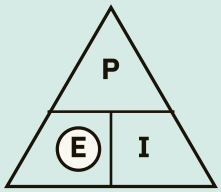
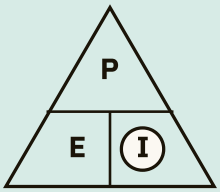
Side by side means multiply. One over the other means divide.

OHM'S LAW

Voltage	Current	Resistance
		
E = I × R	I = E / R	R = E / I

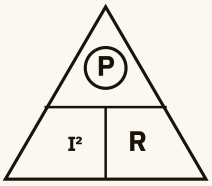
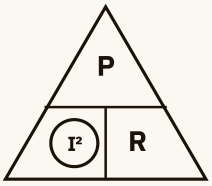
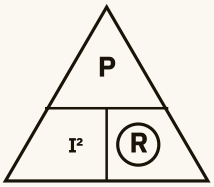
Same numbers as the book: 90 volts and 3 amps. Cover R. 90 / 3 = 30 ohms.

POWER

Power	Voltage	Current
		
P = E × I	E = P / I	I = P / E

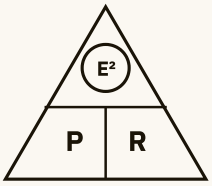
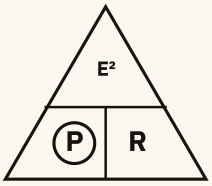
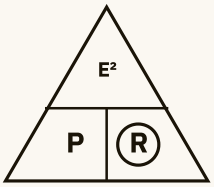
13.8 V × 10 A = 138 W. 12 V × 2.5 A = 30 W. 120 W at 12 V is 10 A, because I = P / E.

POWER WHEEL · CURRENT SQUARED

Power	Current	Resistance
		
P = I² × R	I = √(P / R)	R = P / I²

If the cell says I², the division gives I². Take the square root to get amps.

POWER WHEEL · VOLTAGE SQUARED

Voltage	Power	Resistance
		
E = √(P × R)	P = E² / R	R = E² / P

E² sits on top because E² = P × R. Square root that to get volts.

Letters. E voltage, volts. I current, amperes. R resistance, ohms. P power, watts.

Prefixes, each step ×1,000. pico, nano, micro, milli, the unit, kilo, mega, giga. 1.5 A = 1,500 mA. 28,400 kHz = 28.4 MHz. 500 mW = 0.5 W. Write MHz and kHz.

Decibels. Twice the power is about +3 dB. Ten times is +10 dB. Down to a quarter is about −6 dB.

Also on this test. Series: same current, voltage splits. Parallel: same voltage, current splits. Capacitance (farad) stores an electric field. Inductance (henry) stores a magnetic field. Impedance is the squeeze against AC, still in ohms. The Technician exam uses the first two blocks. The wheel finishes the set.